

1.	Calculate the following for IP address 203.1.1.0/28 to create a simple network based on IP classes. Determine: Total subnets and hosts, first 4 subnet ranges.
2.	A company has been assigned the IP address 216.21.5.0 and requires 30 hosts per subnet. Find 1. Class 2. Default Mask 3. Bit Borrowed 4. New subnet mask 5. Number of Hosts & Subnets 6. First 2 Subnet Ranges
3.	Given the IP address of the device is 192.168.100.0/26. Find the Total subnets, Total hosts per subnet, and the First 3 subnet address ranges.
4.	Determining the Subnet for Devices A and B: a) Device A: 172.16.17.30/20 b) Device B: 172.16.28.15/20
5.	In block of address, we know the IP address of one host is 182.44.82.16/26, What is the first and last address in this block?
6.	An organization is granted the block 16.0.0.0/8. The administrator wants to create 500 fixed-length subnets. a. Find the subnet mask. b. Find the number of addresses in each subnet. c. Find the first and the last address in the first subnet.
7.	Find default subnet mask, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address for 10.0.0.5/8.
8.	Find default subnet mask, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address for 172.16.5.10/16.
9.	Find default subnet mask, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address for 192.168.10.25/24.
10.	Determine whether these devices are in the same subnet- Show working using network addresses Device A: 192.168.10.25/23 Device B: 192.168.11.200/23
11.	Find class, default subnet mask, subnet mask, network bits, host bits, Network address, first valid IP address, Last valid IP address, Broadcast address for 200.100.50.75/26

12.	Find class, default subnet mask, subnet mask, network bits, host bits, Network address, first valid IP address, Last valid IP address, Broadcast address for 130.4.102.1/22
13.	Find default subnet mask, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address for 172.16.10.100/20.
14.	For give IP 150.10.0.0, subnet for 500 hosts Find: 1. Minimum subnet mask 2. Total subnets possible 3. Hosts per subnet
15.	Calculate the IP address to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address. a) 8.1.4.5/16
16.	Calculate the IP address to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address. b) 130.4.102.1/24
17.	Calculate the following for IP address 199.1.1.1/24 to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address.
18.	Calculate the following for IP address 130.4.102.1/22 to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, 1st valid IP address, last valid IP address, and broadcast address.
19.	Calculate the following for IP address 100.64.55.10/13 to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, new subnet mask, 1st valid IP address, last valid IP address, and broadcast address.
20.	Calculate the following for IP address 10.0.0.65/26 to create a simple network based on IP classes. Find default subnet marks, network bits, host bits, hosts per subnet, no of subnets, subnet number, new subnet mask, 1st valid IP address, last valid IP address, and broadcast address.